CAL HACKS 13.0

Landed

No payment to an account the vendor hasn't confirmed.

$3.05B	74%	72 h
BEC losses reported to FBI IC3 in 2025 (24,768 complaints)	of organizations hit by BEC in 2025 (AFP 2026 survey)	after which recovery odds collapse

The attack, three ways

1 A compromised vendor inbox: DMARC-clean, inside the real invoice thread, “we switched banks.”

2 A lookalike domain or display name, with the same message and a never-seen account.

3 A fooled clerk who types the attacker's account into the vendor record by hand. No email, no AI.

The invoice is real. The vendor is real. Only the destination changed.

What Landed does

Every bank-account change becomes a verification case.

Open invoices follow the request and are blocked until the case is verified.

A randomized sub-dollar probe deposit is sent to the new account.

Only the contact of record's read-back of that amount releases the account, in one database transaction.

After payment, Landed checks the money landed; if not, it prepares the recovery package.

The moment

Re-seed the fixtures (Settings → Re-seed) to create the demo cases; the fraud case replays here.

What the agent can and cannot call

Permissions as enforced in the database, not as prompt instructions.

Capability	Agent (LLM / intake / package actions)	Human (dashboard)
Read a vendor email and PDF, extract a requested account	yes	yes (manual edit)
Open a case and queue the probe deposit	yes (via bankAccounts.createUnverified)	yes
Re-point open invoices to the requested account (held)	yes	yes
Record a stated amount	only trusted channels (voice of record, portal login); email is stored rejected	"Simulate vendor answer" in sim mode, labeled
Verify an account (markVerified)	**no**	no (only the bound confirmation does, atomically)
Pay	**no**	yes, and only to a verified current-version account of that vendor
Exceptional release	**no**	two distinct named approvers + written reason, from NEEDS_HUMAN
Edit contacts or account status	**no**	supersede contact; add human-attested record contact
Draft the recovery package prose	yes, from deterministic fields; every number verified	reviews, signs, calls
Know the probe amount	**no (not in prompts, tools, events, or public queries)**	no (operator reads the bank statement; test-only endpoint is key-gate

Real vs. simulated, and the numbers

Real

- The Convex engine and the eight invariants
- Versioned accounts, HMAC-hashed and AES-256-GCM encrypted
- Append-only, redacted audit log
- Rules and Claude extractors (PDFs as documents)
- Vendor portal login; eval harness; Nessie transfers

Simulated (labeled on screen)

- The bank (an in-database ledger unless Nessie is selected)
- Time (a demo clock drives the same timers)
- Vendor answers (a human types what the vendor sees; voice channel written, not dialed)
- Email authentication results; the vendor's bank statement

91	30/30	21/21	0
unit and invariant tests	eval scenario runs (extractor on and off)	through the Claude extractor	false releases

These are results on scripted cases. They demonstrate behavior on these cases; they do not establish effectiveness against real-world fraud.

Prior art, and what Landed adds

Prior art

Trouvé: hold-by-policy plus an automated call with a yes/no transcript judge.

Trustpair, Eftsure: account validation against networks of previously verified vendors.

Charted and similar: detect suspicious change requests, stop at an alert.

Landed adds

- A number read-back instead of yes/no
- A database release gate bound to vendor and account version
- The landing check after payment
- The recovery package
- An eval that includes the legitimate and AI-off paths

Built with: Convex, Next.js, Claude, Nessie, Vapi + Cartesia (written), Playwright.
Tracks: Hyphenate, Microsoft, Convex, Capital One, Cartesia (conditional), Anthropic.

Team: Meridian Outdoor Co. demo · Run your own scenario